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A Multi-view Dataset for Vietnamese Word-Level Sign Language Recognition

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Abstract

This research introduces VSL400, a multi-view video dataset for isolated word-level recognition of Vietnamese Sign Language (VSL). The dataset contains 74,259 manually annotated video clips covering 400 glosses performed by 28 signers, including deaf student signers and additional trained VSL signers. Each signing instance was recorded simultaneously from three synchronized RGB camera views: front, left, and right. This design captures complementary visual perspectives of sign articulation and supports single-view, cross-view, and multi-view recognition studies. To facilitate reproducible research, we provide a preprocessing pipeline for temporal boundary detection, segmentation, and spatial normalization, together with structured metadata and reference code. Metadata, documentation, and processing code are available from Zenodo, while the de-identified human video files are shared through controlled access governed by a Data Usage Agreement. VSL400 addresses a major resource gap for Vietnamese Sign Language and provides a practical reference dataset for research on isolated sign recognition in a low-resource language setting.

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Background & Summary

Hearing loss is one of the most common disabilities worldwide. The World Health Organization estimates that around 5% of the global population, roughly 430 million people including 34 million children, live with disabling hearing loss and require rehabilitation services, and it projects that this figure will reach 700 million, or one in ten people, by 2050.[1] For many of them, sign language is the primary way they communicate: the World Federation of the Deaf reports that more than 72 million deaf people rely on over 300 distinct sign languages in daily life.[2,3]

Sign language is a full natural language in its own right. Rather than sound, it uses handshape, movement, facial expression, body posture, and the use of space to convey meaning, with the same expressive range as any spoken language. Yet deaf and hard-of-hearing people still face real barriers, because most digital and public services are designed around spoken and written language and offer little support for visual-gestural communication,[4] even though signing is central not only to communication but to the cultural identity of deaf communities.[5] Human interpreters can bridge this gap, but they are in short supply and often too costly for everyday use.

Automatic sign language recognition (SLR) and translation (SLT) aim to narrow this gap. By learning to map visual signing into text or speech, and speech back into a written or signed form, such systems can power live captioning, interactive assistants, and real-time interpretation. Tools like these could meaningfully improve access to education, employment, and public services for deaf communities, and make everyday interaction with hearing people far smoother.

A practical obstacle is that sign languages are not interchangeable. Each has its own vocabulary, grammar, and regional variation, so American Sign Language (ASL), Chinese Sign Language (CSL), and the Arabic, German, Indian, and Korean sign languages are genuinely distinct languages rather than dialects of one system.[6–11] A model or dataset built for one of them cannot simply be reused for another, which makes language-specific data a precondition for progress, and it is precisely here that Vietnamese Sign Language remains underserved.

On the modeling side, SLR research has moved steadily from the early CNN-RNN and 3D-CNN pipelines toward transformer-based, skeleton-based, and multimodal methods. Convolutional-recurrent video models remain a solid baseline for isolated recognition,[12] but they struggle with the long-range temporal dependencies of continuous signing, which newer transformer designs address through multi-scale temporal attention.[13] A parallel line of work abandons raw pixels altogether: keypoint- and skeleton-based models reason over body landmarks, lowering computational cost and reducing dependence on background appearance, and recent keypoint-attention and skeleton networks reach competitive accuracy on this lighter representation.[14,15] Because appearance and motion carry complementary cues, fusion models that combine the two are particularly effective at separating signs that look like,[16] while multilingual studies caution that, since sign languages are community-specific, resources from well-supported languages cannot substitute for Vietnamese data.[17]

The solution of these problems requires technologically inclusive answers because they will both improve accessibility and create linguistic equality which supports social inclusion for deaf people in various cultural settings worldwide.

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Available Sign Language Datasets

Curated datasets have been a major driver of progress in sign language recognition (SLR). Even so, the field is still held back by the limited size, narrow modality coverage, and uneven linguistic diversity of existing resources, especially for underrepresented sign languages.

Several large-scale datasets have become standard reference points. (`https://www.robots.ox.ac.uk/~vgg/research/bsl1k/`) BSL-1K[18] is a British Sign Language collection of over 1,000 hours of broadcast footage, weakly annotated with 1,000 glosses mined through self-supervised techniques; its scale is impressive, but the weak supervision and the absence of temporal segmentation limit how it can be used. The (`https://how2sign.github.io/index.html`) How2Sign dataset[19] takes a different approach, offering more than 80 hours of synchronized multi-view, multimodal American Sign Language (ASL) recordings that combine RGB video, depth, skeletal keypoints, and text transcripts.

For continuous SLR, RWTH-PHOENIX-Weather-2014[20] (`https://www-i6.informatik.rwth-aachen.de/~koller/RWTH-PHOENIX/`) remains the most widely used benchmark. This German Sign Language corpus provides 5,672 training, 540 validation, and 629 test videos recorded at 25 FPS by nine signers over a vocabulary of 1,295 glosses, though it is confined to the weather-forecast domain and offers little visual variety. Other widely used resources include (`https://cvml.ankara.edu.tr/datasets/`) AUTSL,[21] with 38,000 Turkish Sign Language videos across 226 classes, and (`https://ustc-slr.github.io/datasets/2021_csl_daily/`) CSL-Daily,[22] a large Chinese Sign Language corpus annotated with both glosses and text. The (`https://www.robots.ox.ac.uk/~vgg/data/bobsl/bobslv1_2.html`) BBC-Oxford BOBSL dataset,[23] while high in quality and produced by professional signers, is still modest in scale relative to the full expressive range of natural signing.

Table 1 summarizes these datasets alongside the attributes most relevant to benchmark design. One gap stands out clearly: Vietnamese Sign Language has no public dataset at all. VSL400 is built to close that gap, providing a high-quality, multi-view resource for reproducible word-level SLR in a low-resource setting.

Table 1: Comparison of benchmark sign language datasets.

Dataset	Lang	Glosses	Clips / Hours	Signers	Modality	View
BSL-1K	BSL	∼1,000	∼1,000 hrs	N/A	RGB (TV)	Frontal
How2Sign	ASL	∼16,000	80 hrs	∼10	RGB, Depth, Pose	Multi
Phoenix-2014	DGS	1,295	7,000+ videos	9	RGB	Frontal
AUTSL	TSL	226	38,000 videos	43	RGB, Depth	Frontal
CSL-Daily	CSL	∼500	50,000+ videos	50	RGB, Text	Frontal
BOBSL	BSL	∼1,000	∼1,400 hrs	N/A	RGB	Frontal
VSL400 (Ours)	VSL	400	74,259 videos	28	RGB	Multi

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Automatic Sign Language Recognition

Automatic sign language recognition (SLR) has been studied for several decades using a wide range of machine learning methods. Before deep learning, SLR pipelines relied on hand-crafted visual features fed into sequential models such as Hidden Markov Models (HMMs) and Conditional Random Fields (CRFs) to capture the temporal structure of signing,[24–26] with classifiers such as Support Vector Machines (SVMs) mapping these features to predefined gloss labels.

Deep learning reshaped the field over the past decade. End-to-end trainable frameworks combined CNNs for spatial feature extraction with RNNs and attention mechanisms for temporal modeling, and delivered substantial accuracy gains across standard benchmarks. Contemporary deep-learning SLR methods fall broadly into RGB-based (visual), skeleton-based, and multimodal approaches.

RGB-based methods learn directly from video, jointly modeling spatial appearance and temporal motion. Early models such as C3D[27] and Inflated 3D ConvNets (I3D)[28] used 3D convolutions to capture motion, and performed strongly when trained on large datasets like MS-ASL[29] and WLASL.[30] Later work introduced sign-specific inductive biases; for example, hand-pose-guided 3D pooling localizes feature aggregation around the hands to sharpen discrimination between visually similar signs.[31] More recently, transformer architectures have set the state of the art in video understanding: the Video Swin Transformer[32] uses hierarchical shifted-window attention to capture both local and global dependencies, while VideoMAE[33] applies masked video modeling with high masking ratios to learn from unlabeled video through spatiotemporal reconstruction.

Skeleton-based methods instead represent the signer as landmark coordinates extracted from each frame, which suppresses background noise and lowers computational cost. Pose is typically estimated with models such as OpenPose,[34] HRNet,[35] or BlazePose.[36] The resulting joint sequences are then modeled as spatio-temporal graphs: ST-GCN[37] connects joints with both spatial and temporal edges, and MS-G3D[38] extends this with multi-scale graph convolutions and gated G3D operators for long-range dependencies. Transformer-based variants such as SPOTER[39] apply self-attention directly to landmark sequences, while hybrid designs pair GCNs with temporal transformers.[41]

Multimodal systems combine these signals. SAM-SLR[42] fuses RGB, depth, and skeletal streams through sign-specific SL-GCN and SSTCN components, while DSTA-SLR[43] improves robustness by dynamically aggregating spatio-temporal relations and multi-scale temporal representations. Such fusion approaches tend to generalize better, particularly on large-scale benchmarks like WLASL.[30]

Vietnamese Sign Language

Vietnamese Sign Language (VSL) has several grammatical features that give it a distinctive character, most notably its heavy reliance on ellipsis and focus-driven emphasis. Ellipsis here refers to the systematic omission of pragmatically recoverable elements-null arguments, semantically light function words, and auxiliaries-which VSL's well-developed spatial indexing and classifier mechanisms make possible. Focus-driven emphasis, in turn, is expressed through non-manual signals such as eyebrow movement, head tilts, and mouthing, prosodic

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cues such as holds and segmental lengthening, and flexible constituent ordering. Together these produce a discourse-adaptive syntax in which functional elements are dropped unless they add substantially to propositional meaning.

An estimated 2.5 million deaf or hard-of-hearing people in Vietnam rely on VSL for everyday communication. Despite the size and importance of this community, the technological resources available for VSL remain scarce, and the lack of reliable sign language recognition (SLR) tools deepens communication inequities and limits full participation in education, employment, and public life.

The distinctiveness of VSL has been acknowledged linguistically, yet it is still largely absent from computational research, especially data-driven work. Existing attempts to build automatic VSL recognition systems[44–46] have been held back by the absence of large-scale, high-quality public datasets, which constrains not only model accuracy and generalization but also the inclusion of VSL in international sign language technology research.

Table 2 contrasts representative prior VSL studies with VSL400 across task, modality, viewpoint coverage, and data availability. All of these earlier efforts are single-view and none released a large-scale word-level corpus. In contrast, VSL400 provides a multi-view, word-level dataset of substantial size, captured by three synchronized cameras; to the best of our knowledge, it is the first multi-view, large-scale, word-level VSL corpus released for research.

Table 2: Representative prior efforts on Vietnamese Sign Language (VSL) recognition compared with VSL400.

Study	**Task**	**Modality**	**Views**	**Public**
Khuat et al. (2021)[44]	Isolated (detection)	RGB + MediaPipe pose	Single	No
Tran et al. (2023)[46]	Continuous SLR	RGB + MediaPipe pose	Single	No
Nguyen et al. (2024)[45]	Isolated word	RGB + depth (CNN)	Single	No
VSL400 (ours)	Isolated word	RGB	Three	Yes

Contributions of This Study

Sign language recognition (SLR) is usually divided into two main tasks: word-level recognition,[47,48] which identifies individual sign glosses, and sentence-level recognition,[49–51] which interprets continuous sign sequences. This study addresses the former, since word-level recognition provides the foundation on which more complex continuous and translation models are built.

Although word-level datasets have grown in number, most are recorded from a single frontal view. Sign language is inherently three-dimensional and context-dependent, so a sign can look very different from another angle, and models trained on frontal data alone tend to generalize poorly to other viewpoints. A frontal view can also make distinct glosses appear visually similar, making them harder to tell apart. Capturing each sign from several viewpoints-frontal, left, and right-addresses these issues by better reflecting real-world conditions and improving robustness.

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This study introduces VSL400, the first controlled-access research benchmark for word-level Vietnamese Sign Language (VSL) recognition, designed to support reproducible, data-driven research in a low-resource setting. Its main contributions are as follows.

First, we release a large-scale, manually annotated word-level VSL dataset of 74,259 video clips covering 400 glosses, recorded from 28 signers (deaf native users and trained hearing signers). It fills a clear gap in publicly accessible VSL resources and provides a foundational corpus for isolated sign recognition.

Second, the data are captured with a synchronized tri-view protocol that records the frontal, left, and right perspectives of each sign instance simultaneously. This design reflects complementary viewpoints of manual articulation and enables systematic study of cross-view variability, viewpoint robustness, and multi-view learning.

Third, we provide a reproducible preprocessing and segmentation pipeline that turns continuous multi-view recordings into standardized, gloss-level clips. It combines pose-based temporal boundary localization with boundary-guided segmentation and spatial normalization, and is released with the dataset to support transparent reuse and fair benchmarking.

Fourth, we supply structured metadata, a standardized data organization, and thorough quality-control procedures-multi-pass manual annotation, inter-annotator agreement checks, and cross-view synchronization verification-which together ensure the consistency, reliability, and long-term usability of the release.

Finally, we report reference baselines from five representative deep learning models spanning RGB-based and skeleton-based recognition paradigms. These serve as technical validation and reproducible reference points rather than an exhaustive performance comparison, providing a starting point for future methodological work built on VSL400.

Methods

Ethics Approval

The VSL400 dataset was collected within the state-funded research project "Application of AI Technology for Converting Sign Language into Speech to Support Hearing-Impaired Students in Communication Activities in Lam Dong Province" (project registration No. 2025-24/KQNC-CS). This number is a research project (governance) registration and is not an ethics committee reference. Data were collected at the Center for Supporting Inclusive Education of Lam Dong Province (formerly the Lam Dong Provincial School for the Deaf) and at Da Lat College of Education, Vietnam.

The institutions involved do not operate a separate formal institutional review board (IRB); in accordance with national practice, ethical oversight and approval were provided by the responsible institutional authorities. Initial written institutional approval to conduct the data collection was obtained in November 2024. Institutional approval to publicly release and share the de-identified VSL400 dataset for controlled research access with domestic and international research groups was subsequently granted and is documented by a signed and sealed permission issued by the Director of the Center for Supporting Inclusive Education of Lam Dong Province and the Rector of Da Lat College of Education.

A total of 28 participants took part in the study: 23 deaf students, who are minors and

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native users of Vietnamese Sign Language (VSL), and 5 additional hearing signers trained in VSL. All adult participants provided written informed consent for both participation in the data collection and controlled sharing of the resulting de-identified dataset. For the deaf participants under 18 years of age, written consent was obtained from a parent or legal guardian together with the participant's assent, in accordance with the approved study protocol and institutional guidelines. The guardian consent covered both participation in the recording sessions and controlled sharing of the de-identified video data and associated metadata for research and development use under the VSL400 Data Usage Agreement. The data are not released as unrestricted public downloads; access is managed through controlled access to reduce privacy and re-identification risks.

Prior to release, all videos were processed using automated facial blurring, and direct personal identifiers were removed from the accompanying metadata. The released metadata do not contain participant names, contact information, school identifiers, or other direct personal identifiers. However, because the videos contain human signing performances that may retain residual biometric and behavioral information, the video files are shared under controlled access governed by a Data Usage Agreement. Data collection, processing, and publication were conducted in compliance with the approved project protocol (No. 2025-24/KQNC-CS) and applicable national research governance regulations.

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Description of VSL400

This section outlines the procedures used for data acquisition, preprocessing, annotation, and segmentation of video recordings into structured gloss-level units. The complete dataset collection and processing pipeline is illustrated in Figure 1.

To construct the dataset, tri-view continuous video recordings of sign sequences were captured using synchronized RGB cameras placed at frontal, left, and right viewpoints. Each session involved a signer performing a predefined set of glosses in a controlled indoor environment with consistent background and lighting conditions.

The raw video files were then subjected to a normalization process to ensure temporal alignment across the three views and standardize encoding parameters such as frame rate, resolution. Following synchronization, automatic segmentation was performed using keypoint-based boundary detection with the Mediapipe framework. This process enabled identification of sign boundaries and extraction of individual word-level clips.

Subsequently, a semi-automatic annotation procedure was employed. Candidate clips were imported into the Label Studio platform, where trained annotators manually verified gloss labels, ensured segmentation quality, and corrected any misalignments or mislabeled instances. The resulting dataset includes high-quality, manually validated samples suitable for reproducible training and benchmarking of sign language recognition systems.



Figure 1: Dataset collection pipeline.

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Participant Recruitment and Data Collection Protocol

The dataset draws on recordings from 28 participants in two groups. Most of the data comes from 23 deaf students at specialized schools for the hearing-impaired, who are native signers of Vietnamese Sign Language. We recorded them in a studio with fixed cameras, consistent lighting, and a green-screen backdrop to keep the recording conditions stable. The remaining five participants were hearing volunteers whom we trained until they could sign the full vocabulary fluently, with deaf experts checking their signing throughout. We filmed this second group outside the studio, in classrooms and homes against plain white and grey walls, so that the dataset would include some variation in signers and backgrounds rather than a single uniform setup.

The dataset needs more diverse visual content which includes different signing styles and various settings to achieve better results. The researchers recorded five additional hearing participants inside a classroom setting which provided them with less controlled recording conditions. The training process for these people involved them to create VSL glosses through the same recording method while deaf experts provided them with assistance. The addition of this group segment brings average diversity to both signer profiles and recording environments which makes the dataset more suitable for real-world modeling applications.

All participants performed a predefined vocabulary of 400 glosses, recorded from three synchronized camera views (frontal, left, and right) at a resolution of 1920×1080 pixels. The original recordings were captured at 30 frames per second and were subsequently standardized to a uniform 25 frames per second during preprocessing using standard FFmpeg frame-rate conversion, prior to all downstream processing. The 25 FPS standard was adopted to enforce a single temporal scale across recordings and to align with widely used sign language recognition benchmarks (e.g., RWTH-PHOENIX-Weather, 25 FPS). MediaPipe pose extraction, temporal boundary localization, and segmentation were all performed on the finalized 25 FPS videos, so that the released videos, metadata, and annotations are mutually consistent. The participants received instructions to look directly at the cameras while keeping their body position stable throughout the signing process for better cross-view alignment and pose estimation.

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Gloss Selection and Vocabulary Construction

The VSL400 vocabulary was designed to reflect high frequency and semantically diverse concepts used in daily communication. Glosses were selected and reviewed by expert annotators fluent in VSL and knowledgeable in linguistic annotation. The vocabulary spans four semantic domains: People and Society, Daily Life, Nature and Environment, and Activities and Concepts. This structured approach supports future domain-specific evaluations and curriculum-based modeling.

A core innovation of our data collection protocol is a three-camera configuration with sensors positioned on the left ($\sim 45°$), Front ($\sim 0°$), and Right ($\sim 45°$) as shown in Fig. 2. This arrangement, comparatively uncommon in existing word-level Sign Language Recognition (SLR) datasets, captures complementary multi-view observations of each sign, thereby enriching the dataset. The resulting diversity of viewpoints enhances its utility for machine learning and computer vision research, enabling more accurate and robust recognition

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models.



Figure 2: Tri-camera VSL data acquisition pipeline: synchronized left–front–right views, fixed camera geometry, controlled lighting, and standardized signer enrollment. The depicted participant (and, for minors, their parents or guardians) provided written consent for the publication of these images.

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Video Preprocessing and Segmentation

To convert raw continuous video recordings into structured and normalized gloss-level clips suitable for training, annotation, and evaluation, we used a semi-automatic preprocessing pipeline consisting of Temporal Boundary Localization (TBL), Boundary-Guided Segmentation and Pruning (BGSP), and human quality-control review. The processing steps are visualized in Figure 1, and the corresponding pseudocode is provided in Algorithm 1 and Algorithm 2.

In the first stage, TBL (Algorithm 1) localises candidate gesture intervals from arm dynamics. For every frame, MediaPipe yields the image-plane coordinates and visibility scores of the shoulder, elbow, and wrist of both arms, and we evaluate each elbow angle against the extension threshold $\theta = 160°$: an arm is regarded as raised when its elbow angle falls below θ and its wrist visibility exceeds 0.6. Importantly, the two arms are followed by independent state machines rather than collapsed into a single aggregate signal. Each machine commits a transition only after the raised, or lowered, condition has persisted for N consecutive frames, which suppresses transient detections and brief tracking dropouts. The onset of a gesture is taken as the instant the *first* arm becomes active, and its offset as the instant the *last* active arm returns to rest; the interval reported for a sign is therefore the union of the active spans of the two arms. This union rule makes boundary localisation agnostic to which hand is dominant and handles one- and two-handed signs without any

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special casing. A symmetric temporal buffer τ_b is finally added on either side of each interval to preserve the preparatory and retraction phases of the gesture.

In the second stage, BGSP (Algorithm 2) transforms each localised interval into a normalised gloss-level clip. Because the persistence requirement enforced during TBL already discards transient, sub-gesture activations, BGSP does not apply a separate duration filter or merging heuristic; the pruning of spurious activations is inherited from that persistence test, and BGSP operates directly on the intervals returned by Algorithm 1. For each interval, the corresponding frame range is extracted and every frame is cropped from the native 1920×1080 recording to a fixed 1080×1080 window centred on the signer, using a horizontal offset $x_{\text{offset}} = 420$ pixels (with $y = 0$) that was fixed empirically for our studio geometry. The cropped frames are re-encoded at the dataset frame rate and indexed by gloss label for subsequent annotation and model training. In the small number of cases where automatic localisation yielded an unusable clip, the segment was corrected or discarded during the manual quality-control stage described below rather than by an automated rule.

The system provides a method to perform segmentation of sign language data which produces results that are both reproducible and have consistent resolution and semantic accuracy for model training and benchmark evaluation.

Algorithm 1 summarises Temporal Boundary Localisation (TBL), which localises candidate gesture intervals from the elbow dynamics of both arms and reports their union.

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Algorithm 1: Temporal Boundary Localization (TBL).

Input: $F = \{f_1, \ldots, f_T\}$ with timestamps t_i `// video frames`
$\theta = 160^\circ$ `// elbow extension threshold`
$\tau_v = 0.6$ `// wrist visibility threshold`
N `// onset/offset persistence in frames`
τ_b `// symmetric temporal buffer, e.g., 0.4 s`
Output: S = list of candidate gesture intervals $[start, end]$

$S \leftarrow \emptyset$;
foreach *arm* $a \in \{\mathrm{L}, \mathrm{R}\}$ **do**
  $state_a \leftarrow \textsc{down}$; $u_a \leftarrow 0$; $d_a \leftarrow 0$; $onset_a, offset_a \leftarrow \perp$;
foreach *frame* f_i *with timestamp* t_i **do**
  Extract pose landmarks of both arms via MediaPipe Pose with landmark smoothing;
  if *pose landmarks are not available* **then**
    skip frame f_i;
    continue;
  foreach *arm* $a \in \{\mathrm{L}, \mathrm{R}\}$ **do**
    Extract $\mathrm{shoulder}_a$, elbow_a, and wrist_a;
    $\alpha_a \leftarrow \angle(\mathrm{shoulder}_a, \mathrm{elbow}_a, \mathrm{wrist}_a)$;
    $v_a \leftarrow \mathrm{vis}(\mathrm{wrist}_a)$;
    $reliable_a \leftarrow (v_a \geq \tau_v)$;
    $active_a \leftarrow reliable_a \wedge (\alpha_a < \theta)$;
    `// Low-visibility wrist landmarks are not used as reliable active-state evidence`
    if $state_a = \textsc{down}$ **then**
      if $active_a$ **then**
        if $u_a = 0$ **then**
          $t_a^{\mathrm{on}} \leftarrow t_i - \tau_b$;
        $u_a \leftarrow u_a + 1$;
        if $u_a \geq N$ **then**
          $state_a \leftarrow \textsc{up}$; $onset_a \leftarrow t_a^{\mathrm{on}}$; $u_a \leftarrow 0$;
      else
        $u_a \leftarrow 0$;
        $t_a^{\mathrm{on}} \leftarrow \perp$;
        `// Reset pending onset`
    else
      if $active_a$ **then**
        $d_a \leftarrow 0$;
        $t_a^{\mathrm{off}} \leftarrow \perp$;
        `// Reset pending offset`
      else
        if $d_a = 0$ **then**
          $t_a^{\mathrm{off}} \leftarrow t_i + \tau_b$;
        $d_a \leftarrow d_a + 1$;
        if $d_a \geq N$ **then**
          $state_a \leftarrow \textsc{down}$; $offset_a \leftarrow t_a^{\mathrm{off}}$; $d_a \leftarrow 0$;

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The function UNION returns a completed interval only after every arm that became active has returned to rest. If a single arm was used during the sign, its own interval $[onset, offset]$ is reported. If both arms were used, the reported interval is

$$[\min(onset_{\mathrm{L}}, onset_{\mathrm{R}}), \max(offset_{\mathrm{L}}, offset_{\mathrm{R}})].$$

In all other configurations, the interval remains undefined and frame processing continues.

The current implementation of the TBL code does not attempt to interpolate missing MediaPipe pose landmarks. As such, frames that do not contain any pose landmarks for a given hand are simply skipped. Also, it is not considered reliable active-state evidence for candidate boundaries the low-visibility wrist landmarks. While the short tracking persistence requirements for the onset and offset of a given active state help to mitigate the effects of discontinuities in the pose tracking for a given hand, longer occlusions, severe motion blur, and other similar factors can still affect the quality of candidate boundaries. For this reason, the TBL is used here as a candidate boundary generator, rather than as a fully automatic annotator. Candidate clips for the generated boundaries are reviewed by a human for quality control before they are included in the dataset that will be released.

Algorithm 2 summarises Boundary-Guided Segmentation and Pruning (BGSP), which trims each localised interval to its boundaries and applies a fixed spatial crop to generate normalised 1080×1080 clips.

Algorithm 2: Boundary-Guided Segmentation and Pruning (BGSP).

Input: S `// gesture intervals from Algorithm 1`
$W \times H = 1920 \times 1080$ `// native resolution`
$(w, h, x, y) = (1080, 1080, 420, 0)$ `// fixed crop window`
fps `// dataset frame rate (25)`
Output: $Clips$ = list of normalized gloss-level clips
$Clips \leftarrow \emptyset$;
foreach *interval* $[start, end] \in S$ **do**
  $start_frame \leftarrow \lfloor start \times fps \rfloor$; $end_frame \leftarrow \lfloor end \times fps \rfloor$;
  $Buffer \leftarrow \emptyset$;
  for $i \leftarrow start_frame$ **to** end_frame **do**
    $f' \leftarrow$ crop f_i to window $[x : x + w,\ y : y + h]$; `//` $1920 \times 1080 \rightarrow 1080 \times 1080$
    Append f' to $Buffer$;
  $Clip \leftarrow$ encode $Buffer$ at fps;
  Assign a unique identifier and gloss label to $Clip$; save $Clip$;
  Append $Clip$ to $Clips$;
return $Clips$

All segmented clips were saved with consistent naming conventions, unique identifiers, and metadata files specifying gloss label, signer ID, and recording parameters. Both algorithms are implemented in Python using OpenCV and MediaPipe. The code and configuration scripts are publicly available (see Code Availability section).

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Annotation and Quality Control

Gloss annotations were manually verified by trained sign linguists through a two-pass validation process. Each clip was reviewed for signing accuracy, gloss alignment, and visual clarity. Annotators were provided with recording protocols and domain, specific gloss lists to ensure consistency. Discrepant cases were adjudicated jointly to reach consensus. Annotation consistency exceeded 95% agreement across the corpus.

Data records

The VSL400 dataset described in this Data Descriptor is deposited in Zenodo[40] under DOI: `https://doi.org/10.5281/zenodo.17943574`. The record provides metadata, documentation, processing code, and controlled access to the de-identified human video files. The video archive is organized into three camera folders named `front_view`, `left_view`, and `right_view`, each containing gloss-level sign clips from one synchronized camera view. Videos with the same zero-padded six-digit identifier across the three folders correspond to the same signing instance recorded simultaneously from the three viewpoints.

The top-level directory structure is organized as follows:

```
VSL400/
|-- front_view/
|    |-- 000000.mp4
|    |-- 000001.mp4
|    `-- ...
|-- left_view/
|-- right_view/
|-- front_view.json
|-- left_view.json
`-- right_view.json
```

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All videos are encoded as MP4 with H.264 codec. They are all rendered at 25 frames per second and are cropped and trimmed to a fixed spatial resolution of 1080×1080 pixels.

The metadata for each view (front_view.json, left_view.json, right_view.json) is the annotation of the videos. The metadata of each clip of Figure 3 shows the video ID, the anonymized ID of the signer, the gloss label, the frame number and the runtime of the clip in seconds. As the structure of the metadata files for the three views is the same, the videos can be linked across the views based on the video_id field. The released metadata does not contain participant names, contact information, school identifiers, or other direct personal identifiers. The videos were de-identified by automated facial blurring and removal of direct identifiers.

All clips are temporally segmented, spatially normalized and indexed consistently across views to simplify benchmarking and reproducible evaluation.



Figure 3: Example of a JSON metadata entry.

Data Overview

The final VSL400 release contains 74,259 annotated gloss-level video clips covering 400 glosses and 28 signers. Each sign instance was recorded simultaneously from three synchronized views: `front_view`, `left_view`, and `right_view`.

The processed collection comprises approximately 53.99 hours of video, with an average clip duration of 2.61 seconds and a compressed size of 67.3 GB. Only processed and segmented gloss-level clips are included in the release; the raw continuous recordings are not shared because of privacy and data-governance requirements. Table 3 summarizes the main dataset statistics.

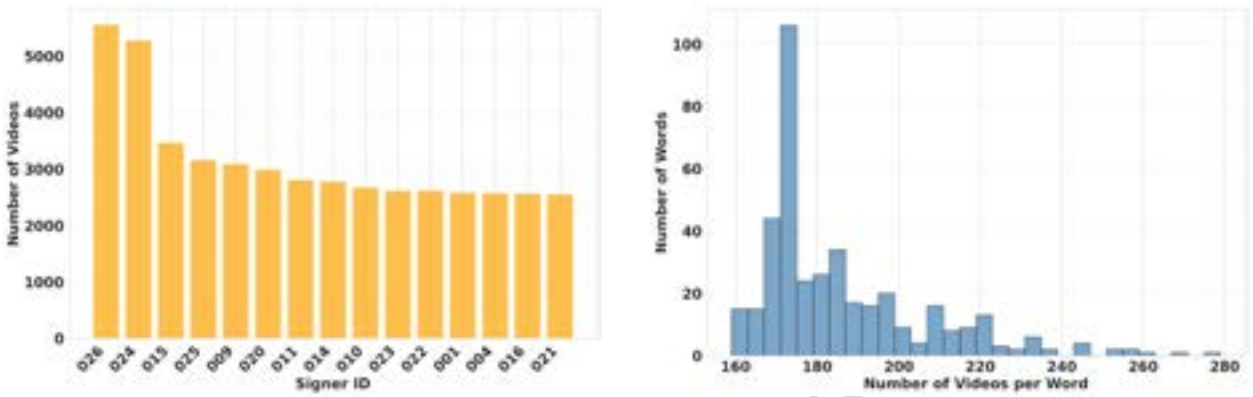
Table 3: Statistics of the VSL400 Dataset.

Attribute	**Value**	**Notes**
Number of glosses	400	Annotated manually
Total videos	74,259	Across all views
Average videos per gloss	185.6	Averaged over 400 glosses
Number of signers	28	23 deaf students, 5 hearing signers
Recording views	3	Frontal, Left, Right
Clip duration (avg)	2.61 s	After temporal segmentation
Total duration	53.99 hours	Normalized format

To help users assess the composition of the corpus, Figure 4 summarizes two descriptive

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properties of VSL400: signer-level contribution and gloss-level sample distribution. These overview plots show that the dataset does not depend disproportionately on a single signer or gloss.



(a) Number of samples contributed by the most active signers.

(b) Distribution of the number of samples per gloss.

Figure 4: Overview of VSL400 composition. (a) Signer-level sample contribution. (b) Gloss-level sample distribution across the 400-word vocabulary.

Technical Validation

To ensure the reliability and quality of the VSL400 dataset, we implemented a rigorous multi-stage validation process encompassing both data integrity verification and systematic performance benchmarking. The validation procedure consisted of two primary components: (1) comprehensive quality control checks to verify annotation accuracy and temporal consistency across multi-view recordings, and (2) extensive benchmarking experiments using state-of-the-art models in both RGB and pose-based sign language recognition. These evaluations not only quantify baseline model performance under controlled protocols but also serve as empirical evidence of the dataset's technical robustness and utility for advancing sign language recognition research.

Data Quality Assurance

Prior to model evaluation, we conducted a systematic two-stage quality control process to validate all annotations and ensure temporal consistency across the multi-view recordings.

Length Verification

To identify potential annotation errors and redundant content, we performed statistical analysis of video durations across all samples. Violin plots were generated for each sign class to visualize the distribution of video lengths and detect outliers-videos that were unusually short or excessively long compared to the class median. Videos flagged as outliers underwent manual re-examination to determine whether they contained redundant actions, incomplete

gestures, or unnecessary pre/post-sign movements. Videos confirmed to have redundant content were precisely trimmed to retain only the core sign execution, ensuring temporal consistency across the dataset. The violin plot, as shown in Figure 5, highlights the distribution of video lengths for a representative sign class. The kernel density estimate shows where the samples are concentrated around the median length, while the individual dots emphasize probable outliers.

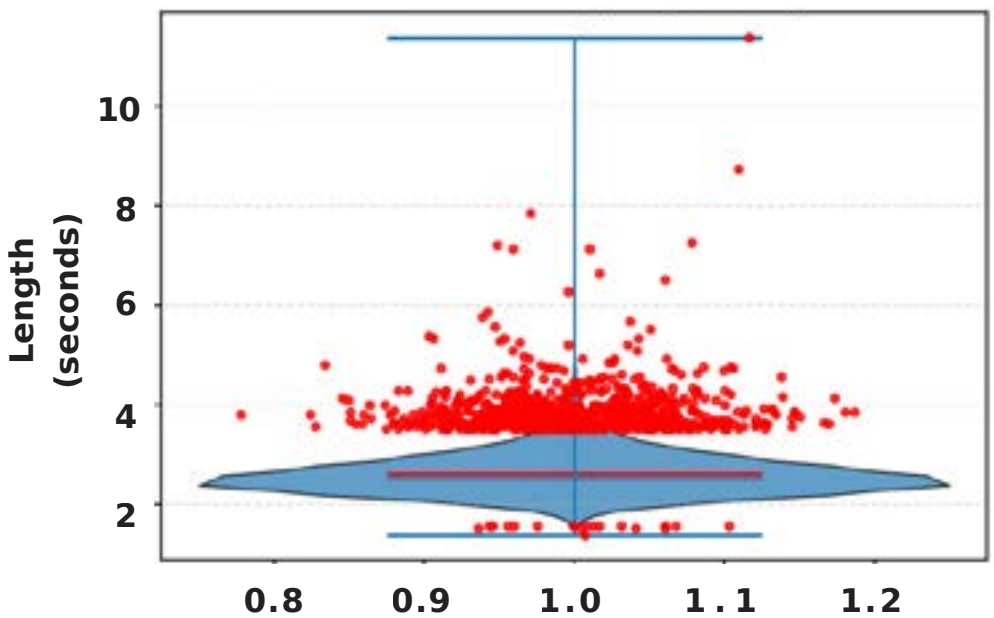


Figure 5: Violin plot of video length distribution for duration-based validation.

Multi-View Synchronization Verification

To verify annotation accuracy and temporal alignment across viewpoints, we implemented a synchronized verification protocol. For each sign instance, the three corresponding videos (front, left, and right views) were displayed simultaneously alongside their annotated labels in a custom verification interface. As illustrated in Figure 6, this interface allowed human validators to assess each sign from multiple perspectives in real time. Human reviewers checked that: (1) all three videos captured the same sign performance, (2) the temporal alignment across views was maintained, (3) the assigned labels were correct for all viewpoints, and (4) no view-specific recording artifacts or synchronization drift existed. Any discrepancies identified during this process triggered re-annotation or re-recording of the affected samples to maintain dataset integrity.

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Figure 6: Example of simultaneous cross-view validation. All depicted participants (and, for minors, their parents or guardians) provided written consent for the publication of these images.

Model Selection and Implementation

For technical validation of the VSL400 dataset, we report reference baselines using five representative deep learning models that cover the major paradigms commonly adopted in vision-based sign language recognition. The objective of this evaluation is not to exhaustively optimize recognition performance, but rather to provide reproducible reference points that help assess dataset consistency and illustrate the impact of the proposed tri-view recording design across different modeling assumptions.

- **RGB-based model (VideoMAE)**:[33] a masked autoencoder for video that operates directly on spatiotemporal RGB input and has demonstrated strong performance on large-scale SLR and action-recognition benchmarks. In our setting, VideoMAE is fine-tuned on VSL400 clips using frames from the front view or from all three synchronized views.

- **RGB-based model (Swin3D)**:[32] a Video Swin Transformer backbone for spatiotemporal visual understanding that uses hierarchical shifted-window attention to capture both local motion cues and long-range dependencies in video. We adopt the torchvision swin3d_s implementation (Kinetics-400 pretrained); for brevity, this model is referred to as Swin3D throughout the manuscript and the Supplementary tables. It is included as an additional RGB reference to examine whether the benefits of the multi-view design generalize across transformer-based video architectures.

- **Skeleton-aware (SAM-SLR/SL-GCN baseline)**:[42] a skeleton-aware multimodal recognition framework that integrates complementary visual appearance and pose-based cues through modality-specific encoders and a fusion mechanism. This model is used to evaluate how synchronized multi-view recordings support multimodal feature learning. Because VSL400 is recorded in RGB only and contains no depth modality, we adopt the skeleton branch of this framework — the SL-GCN bone stream operating

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on pose landmarks extracted from each camera view — rather than the complete multimodal pipeline.

- **Skeleton-aware model (DSTA-SLR)**:[43] a skeleton-aware approach that emphasizes dynamic spatial-temporal aggregation to better capture articulation patterns and temporal relations in sign language sequences. On VSL400, DSTA-SLR serves as a representative skeleton-aware baseline to assess the robustness of structured motion modeling under different camera viewpoints.
- **Skeleton-based model (SPOTER)**:[39] a sign pose-based transformer that consumes sequences of 2D body landmarks and is designed to be efficient and robust to signer variation. On VSL400, SPOTER is trained on pose sequences extracted independently from each camera view and then evaluated in both single-view and tri-view regimes.

Across all baselines, we adopt the respective official implementations with minimal modifications and apply a unified evaluation protocol, including signer-disjoint data splits, consistent clip normalization, and matched training and evaluation settings. This design choice ensures that the reported results primarily reflect the intrinsic properties of the VSL400 dataset and the impact of the tri-view setup, rather than architecture-specific optimization.

Front-view and tri-view baseline validation

To assess whether VSL400 provides a consistent and usable basis for recognition research, we evaluated the reference models under two simple data-use settings: training with the front view only and training with all three synchronized views. The dataset was divided into training, validation, and test subsets using an 80/10/10 signer-disjoint protocol.

The results reported in Table 4 were obtained under a 250-epoch validation setting. The purpose of this experiment is not to establish a new state-of-the-art model comparison, but to verify that the released data can support stable training and meaningful evaluation across different recognition paradigms. The same epoch setting is used in the supplementary view-level analyses so that the main manuscript and Supplementary Information follow a consistent validation protocol. Detailed training configurations, hardware information, and additional view-level analyses are provided in the Supplementary Information.

Table 4: Technical validation of VSL400 under front-view and tri-view training configurations.

Model	**Front-view** (Top-1 / Top-5)	**Tri-view** (Top-1 / Top-5)	**Gain** (Top-1 / Top-5)
VideoMAE	73.13 / 83.23	**75.28 / 86.01**	+2.15 / +2.78
Swin3D	76.00 / 90.12	**78.24 / 92.93**	+2.24 / +2.81
DSTA-SLR	80.14 / 90.34	**83.34 / 94.26**	+3.20 / +3.92
SAM-SLR	85.35 / 93.62	**89.60 / 98.31**	+4.25 / +4.69
SPOTER	83.63 / 92.12	**88.26 / 96.26**	+4.63 / +4.14

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Table 4 shows that all evaluated models can learn from the released clips and produce consistent recognition performance on the held-out test set. The improvement observed when using all three views indicates that the synchronized multi-view design contributes complementary visual information beyond the frontal view alone. These results therefore serve as technical validation of the dataset quality, annotation consistency, and multi-view organization, rather than as an optimized ranking of recognition architectures.

Additional cross-view, missing-view, training-configuration, and confusion-matrix analyses are provided in the Supplementary Information. These analyses are included to help future users understand the practical behavior of the dataset under different view-availability conditions.

Usage Notes

Multi-view data and 3D reconstruction

VSL400 contains synchronized RGB videos captured from three viewpoints: frontal, left-oblique, and right-oblique. Clips that share the same zero-padded identifier in the three view folders refer to the same signing instance. This structure is useful for recognition-oriented studies, including single-view evaluation, cross-view testing, multi-view fusion, and missing-view experiments.

The dataset, however, was not designed as a calibrated 3D reconstruction resource. During recording, the camera placement was selected to provide three complementary visual perspectives of the signer. We did not perform a formal camera-calibration procedure with a checkerboard, Charuco board, or other calibration target. For this reason, the release does not include intrinsic matrices, extrinsic matrices, or lens-distortion coefficients.

Researchers should therefore avoid using VSL400 for metric 3D triangulation of hand, body, or skeletal motion. Such reconstruction would require calibrated cameras and a defined world coordinate system. The main value of the three-view setup lies in synchronized multi-view recognition rather than precise 3D geometry. If users extract landmarks with MediaPipe or similar tools, these landmarks should be treated as image-space coordinates with relative depth estimates when available, not as calibrated 3D points reconstructed from the three cameras. Nonetheless, the camera geometry was kept fixed and repeatable throughout data collection, which supports view-consistent representation learning and qualitative cross-view analysis; for groups that require metric reconstruction, calibration parameters could be estimated and released in a future version of the dataset.

Temporal boundary uncertainty and padding

The TBL pipeline applies a symmetric temporal buffer around the detected candidate interval. This buffer was selected as a conservative preprocessing setting to reduce the risk of truncating sign onset and offset movements. For short glosses, the buffer may include non-signing or transition frames near the clip boundaries. The reported baseline results should therefore be interpreted under this preprocessing convention.

The present release does not include an ablation study over different temporal-buffer

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values, nor does it apply Dynamic Time Warping or soft temporal alignment before model input. Future users may evaluate alternative values of τ_b, adaptive padding, or sign-duration-dependent trimming when studying the sensitivity of RGB, 3D-CNN, transformer, or pose-based models to boundary frames.

Background variation and shortcut learning

A recognised risk for vision-based sign language recognition is shortcut learning, in which a model lowers its training error by exploiting incidental cues, such as the recording background or a signer's appearance, rather than the articulated gesture itself. In VSL400 the two participant groups were recorded under different conditions: the deaf students, who contribute the majority of the data, were filmed against a uniform green-screen (chroma-key) background, whereas the hearing signers were filmed against plain white and grey walls in both classroom and home settings. Several considerations bear on the shortcut-learning concern. Because the green-screen background is essentially constant across all glosses, it carries no class-discriminative information and cannot, on its own, act as a shortcut for distinguishing one gloss from another; the more pertinent issues are the limited resemblance of a chroma-key scene to cluttered real-world environments and the possibility that chroma artefacts, such as colour spill or cast shadows, are learned as spurious features. These non-green-screen recordings provide a contrasting, more naturalistic background condition, although we note that recording environment is correlated with signer group, so any environment-controlled comparison should account for this confound. The released face blurring further removes a salient identity cue, and the synchronized multi-view structure requires a model to generalise across three viewpoints rather than to memorise a single scene. Importantly, the dataset remains well suited to background-invariant approaches: skeleton- and keypoint-based models, which discard raw pixels entirely,[14] are largely immune to background shortcuts, and we provide skeleton baselines precisely so that users can contrast appearance-based and pose-based behaviour. For studies that rely on RGB input, we recommend standard countermeasures, namely background removal or randomisation together with appearance augmentation, and we encourage reporting recognition accuracy under such perturbations as a robustness check. A systematic background-ablation study, in which models are retrained on background-suppressed inputs and compared against their raw-RGB counterparts, is a valuable direction for future work built on VSL400.

Dataset diversity and potential biases

VSL400 was assembled with several sources of diversity in mind, yet, like any single-site corpus, it carries biases that users should weigh when interpreting results or deploying trained models. On the positive side, the dataset draws on both deaf native signers and hearing signers, captures every gloss from three synchronized viewpoints, and is reasonably balanced across glosses and across the most active signers, which supports signer-disjoint evaluation and reduces the chance that a model overfits to a particular individual or class.

Several limitations nonetheless temper this diversity. First, the signer pool is modest in size (28 individuals) and demographically narrow: the deaf participants are school-age students and the hearing participants are university students, so the corpus is skewed towards

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younger signers and does not represent the full age range of the VSL-using community. Second, all data were collected through a single institutional partner in Lam Dong Province; because VSL exhibits regional and generational lexical variation, models trained on VSL400 may not transfer cleanly to other regional varieties or to older signers without additional data. Third, the corpus is restricted to isolated, word-level signs drawn from a general everyday vocabulary, and therefore does not cover continuous signing, co-articulation between adjacent signs, sentence-level grammar, or specialised and fingerspelled vocabulary. Finally, the released videos are face-blurred to protect participant identity, which simultaneously removes some of the non-manual facial information that certain signs rely on; analyses that depend heavily on facial expression should take this into account.

We make these characteristics explicit so that users can scope their claims accordingly. VSL400 is well suited to studies of multi-view, signer-generalised, word-level recognition under controlled conditions, whereas conclusions about robustness across regions, ages, recording environments, or connected signing should be drawn with care. Expanding the signer pool across additional regions, age groups, and recording settings is an important direction for future versions of the dataset.

Controlled access

VSL400 contains de-identified human signing videos and human-derived visual data distributed under controlled access. Use of these files is governed by the publicly available VSL400 Data Usage Agreement (DUA): `https://zenodo.org/records/21366957`. Prospective users must complete and sign the DUA and submit it to the Data Provider with their name, affiliation, contact email, and a brief description of the intended research or development purpose. Requests are normally reviewed within 10 business days for documentation completeness, identity or contact verification where needed, and agreement to the DUA terms. They are not assessed based on scientific impact, commercial status, nationality, institution type, or project topic. Access is granted subject to compliance with the DUA and applicable legal, ethical, institutional, and repository requirements.

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Data Availability

The VSL400 dataset is deposited in Zenodo under DOI: `https://doi.org/10.5281/zenodo.17943574`. Metadata, documentation, and processing code are provided to support reproducible research. The de-identified human video files are shared through a controlled-access mechanism because they contain human signing performances that may retain residual biometric and behavioral information even after facial blurring and removal of direct identifiers. Access requests are reviewed through Zenodo, and approved users must agree to the accompanying Data Usage Agreement before using the video files.

Code availability

All preprocessing scripts and baseline model implementation code used in this study are deposited in Zenodo under DOI: 10.5281/zenodo.17943574 (Version 1.0). The code release

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includes the TBL and BGSP preprocessing pipeline, data utilities, and baseline training and evaluation scripts. The human video files are subject to the controlled-access conditions described in the Data Availability section.

References

[1] Organization, W. H.. Deafness and hearing loss. Available at: https://www.who.int/news-room/fact-sheets/detail/deafness-and-hearing-loss.

[2] Nations, U.. International Day of Sign Languages. (2017) Available at: https://www.un.org/en/observances/sign-languages-day.

[3] Needs, E. A. f. S., Education, I.. 'Sign Languages are for Everyone!' UN International Day of Sign Languages. (2020) Available at: https://www.european-agency.org/news/sign-languages-are-everyone-un-international-day-sign-languages.

[4] Alyami, S., Luqman, H., Hammoudeh, M.. Reviewing 25 years of continuous sign language recognition research: Advances, challenges, and prospects. *Information Processing & Management* ***61*** *(5)*, 103774 (2024) doi:https://doi.org/10.1016/j.ipm.2024.103774.

[5] Sarkar, D. R., Ghosh, A.. Challenges faced by students with hearing impairment in higher education: A comprehensive analysis. *International Journal of Speech and Audiology* ***5*** *(1)*, 06–12 (2024) doi:10.22271/27103846.2024.v5.i1a.43.

[6] Athitsos, V., Neidle, C., Sclaroff, S., Nash, J., Stefan, A., Yuan, Q., Thangali, A.. The American Sign Language Lexicon Video Dataset. In *2008 IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops*, 1-8 (2008) doi:10.1109/CVPRW.2008.4563181.

[7] Lin, Y., Chai, X., Zhou, Y., Chen, X.. Curve Matching from the View of Manifold for Sign Language Recognition. In *Computer Vision - ACCV 2014 Workshops, Springer International Publishing*, 233–246 (2015)

[8] Shohieb, S. M., Elminir, H. K., Riad, A.. SignsWorld Atlas; a benchmark Arabic Sign Language database. *Journal of King Saud University - Computer and Information Sciences* ***27*** *(1)*, 68-76 (2015) doi:https://doi.org/10.1016/j.jksuci.2014.03.011.

[9] Forster, J., Schmidt, C., Hoyoux, T., Koller, O., Zelle, U., Piater, J., Ney, H.. RWTH-PHOENIX-Weather: A Large Vocabulary Sign Language Recognition and Translation Corpus. In *Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC'12), European Language Resources Association (ELRA)*, 3785–3789 (2012) Available at: https://aclanthology.org/L12-1503/.

[10] Sridhar, A., Ganesan, R. G., Kumar, P., Khapra, M.. INCLUDE: A Large Scale Dataset for Indian Sign Language Recognition. In *Proceedings of the 28th ACM International Conference on Multimedia, Association for Computing Machinery*, 1366–1375 (2020) doi:10.1145/3394171.3413528.

ARTICLE IN PRESS

[11] Yang, S., Jung, S., Kang, H., Kim, C.. The Korean Sign Language Dataset for Action Recognition. In *MultiMedia Modeling, Springer International Publishing*, 532–542 (2020)

[12] Boukdir, A., Benaddy, M., Ellahyani, A., El Meslouhi, O. & Kardouchi, M. Isolated Video-Based Arabic Sign Language Recognition Using Convolutional and Recursive Neural Networks. *Arabian Journal for Science and Engineering* **47**, 2187–2199 (2022). https://doi.org/10.1007/s13369-021-06167-5

[13] Alyami, S. & Luqman, H. Swin-MSTP: Swin Transformer with Multi-Scale Temporal Perception for Continuous Sign Language Recognition. *Neurocomputing* **617**, 129015 (2025). https://doi.org/10.1016/j.neucom.2024.129015

[14] Guan, M., Wang, Y., Ma, G., Liu, J. & Sun, M. MSKA: Multi-Stream Keypoint Attention Network for Sign Language Recognition and Translation. *Pattern Recognition* **165**, 111602 (2025). https://doi.org/10.1016/j.patcog.2025.111602

[15] Renjith, S., Suresh, M. S. S. & Rashmi, M. An Effective Skeleton-Based Approach for Multilingual Sign Language Recognition. *Engineering Applications of Artificial Intelligence* **143**, 109995 (2025). https://doi.org/10.1016/j.engappai.2024.109995

[16] Renjith, S., Varghese, A., Rashmi, M. & S. S., P. Transformer-Based Motion-Visual Integrated Fusion for Isolated Sign Language Recognition. *Computers and Electrical Engineering* **130**, 110902 (2026). https://doi.org/10.1016/j.compeleceng.2025.110902

[17] Najib, F. M. A Multi-Lingual Sign Language Recognition System Using Machine Learning. *Multimedia Tools and Applications* **84**, 27987–28011 (2025). https://doi.org/10.1007/s11042-024-20165-3

[18] Albanie, S. et al. (2020). BSL-1K: Scaling Up Co-articulated Sign Language Recognition Using Mouthing Cues. In: Vedaldi, A., Bischof, H., Brox, T., Frahm, JM. (eds) Computer Vision – ECCV 2020. ECCV 2020. Lecture Notes in Computer Science(), vol 12356. Springer, Cham. https://doi.org/10.1007/978-3-030-58621-8_3

[19] Duarte, A., Giró-i-Nieto, X., Torres, J., Farella, E., others. How2Sign: A Large-Scale Multimodal Dataset for Continuous American Sign Language. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2021)

[20] Koller, O., Forster, J., Ney, H.. Continuous sign language recognition: Towards large vocabulary statistical recognition systems handling multiple signers. *Computer Vision and Image Understanding* ***141***, 108–125 (2015)

[21] Sincan, O. M., Keles, H. Y.. AUTSL: A Large Scale Multi-Modal Turkish Sign Language Dataset and Baseline Methods. *IEEE Access* ***8***, 181340-181355 (2020) doi:10.1109/ACCESS.2020.3028072.

ARTICLE IN PRESS

ARTICLE IN PRESS

[22] Zhou, H., Zhou, W., Li, H., others. Improving Sign Language Translation With Monolingual Data by Sign Back-Translation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (2021)

[23] Zisserman, S. A. A. G. V. A. L. M. A. H. B. A. T. A. A. H. C. A. N. F. A. B. W. A. R. C. A. A. M. A. A.. BBC-Oxford British Sign Language Dataset. *HAL (Le Centre pour la Communication Scientifique Directe)* (2022) Available at: https://hal.science/hal-03516444.

[24] Pansare, J. R., Ingle, M.. Vision-based approach for American Sign Language recognition using Edge Orientation Histogram. In *2016 International Conference on Image, Vision and Computing (ICIVC)*, 86-90 (2016) doi:10.1109/ICIVC.2016.7571278.

[25] Bessa Carneiro, S., De M. Santos, E. D., De A. Barbosa, T. M., Ferreira, J. O., Soares Alcalá, S. G., Da Rocha, A. F.. Static gestures recognition for Brazilian Sign Language with kinect sensor. In *2016 IEEE SENSORS*, 1-3 (2016) doi:10.1109/ICSENS.2016.7808522.

[26] Adeyanju, I., Bello, O., Adegboye, M.. Machine learning methods for sign language recognition: A critical review and analysis. *Intelligent Systems with Applications* ***12***, 200056 (2021) doi:10.1016/j.iswa.2021.200056.

[27] Tran, D., Bourdev, L., Fergus, R., Torresani, L., Paluri, M.. Learning Spatiotemporal Features with 3D Convolutional Networks. In *2015 IEEE International Conference on Computer Vision (ICCV)*, 4489-4497 (2015) doi:10.1109/ICCV.2015.510.

[28] Carreira, J., Zisserman, A., Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 4724-4733 (2017) doi:10.1109/CVPR.2017.502.

[29] Vaezi Joze, H., Koller, O.. MS-ASL: A Large-Scale Data Set and Benchmark for Understanding American Sign Language. In *The British Machine Vision Conference (BMVC)* (2019) Available at: https://www.microsoft.com/en-us/research/publication/ms-asl-a-large-scale-data-set-and-benchmark-for-understanding-american-sign-language/.

[30] Li, D., Opazo, C. R., Yu, X., Li, H.. Word-level Deep Sign Language Recognition from Video: A New Large-scale Dataset and Methods Comparison. In *2020 IEEE Winter Conference on Applications of Computer Vision (WACV), IEEE Computer Society*, 1448-1458 (2020) doi:10.1109/WACV45572.2020.9093512.

[31] Hosain, A. A., Selvam Santhalingam, P., Pathak, P., Rangwala, H., Košecká, J.. Hand Pose Guided 3D Pooling for Word-level Sign Language Recognition. In *2021 IEEE Winter Conference on Applications of Computer Vision (WACV)*, 3428-3438 (2021) doi:10.1109/WACV48630.2021.00347.

[32] Liu, Z., Ning, J., Cao, Y., Wei, Y., Zhang, Z., Lin, S., Hu, H.. Video Swin Transformer. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 3192-3201 (2022) doi:10.1109/CVPR52688.2022.00320.

ARTICLE IN PRESS

ARTICLE IN PRESS

[33] Tong, Z., Song, Y., Wang, J. & Wang, L. VideoMAE: masked autoencoders are data-efficient learners for self-supervised video pre-training. Adv. Neural Inf. Process. Syst. 35, 3487–3501 (2022). https://doi.org/10.5555/3600270.3601002

[34] Cao, Z., Simon, T., Wei, S., Sheikh, Y.. Realtime Multi-person 2D Pose Estimation Using Part Affinity Fields. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 1302-1310 (2017) doi:10.1109/CVPR.2017.143.

[35] Sun, K., Xiao, B., Liu, D., Wang, J.. Deep High-Resolution Representation Learning for Human Pose Estimation. (2019) arXiv:1902.09212. Available at: https://arxiv.org/abs/1902.09212.

[36] Bazarevsky, V., Grishchenko, I., Raveendran, K., Zhu, T., Zhang, F., Grundmann, M.. BlazePose: On-device Real-time Body Pose tracking. (2020) arXiv:2006.10204. Available at: https://arxiv.org/abs/2006.10204.

[37] Yan, S., Xiong, Y., Lin, D.. Spatial temporal graph convolutional networks for skeleton-based action recognition. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence and Thirtieth Innovative Applications of Artificial Intelligence Conference and Eighth AAAI Symposium on Educational Advances in Artificial Intelligence, AAAI Press* (2018)

[38] Liu, Z., Zhang, H., Chen, Z., Wang, Z., Ouyang, W.. Disentangling and Unifying Graph Convolutions for Skeleton-Based Action Recognition. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 140-149 (2020) doi:10.1109/CVPR42600.2020.00022.

[39] Boháček, M., Hrúz, M.. Sign Pose-based Transformer for Word-level Sign Language Recognition. In *2022 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, 182-191 (2022) doi:10.1109/WACVW54805.2022.00024.

[40] Nguyen Quoc, T., Pham Dang, K., Truong Duy, V., Truong Hoang, V., Bilik, S., Sindelar, M., Stefansky, J., Łysiak, A., and Martinek, R., "VSL400: A Multi-view Dataset for Vietnamese Word-Level Sign Language Recognition," Zenodo (2025), doi: 10.5281/zenodo.17943574.

[41] Tunga, A., Nuthalapati, S. V., Wachs, J.. Pose-based Sign Language Recognition using GCN and BERT. (2020) arXiv:2012.00781. Available at: https://arxiv.org/abs/2012.00781.

[42] Jiang, S., Sun, B., Wang, L., Bai, Y., Li, K., Fu, Y.. Skeleton Aware Multi-modal Sign Language Recognition. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 3408-3418 (2021) doi:10.1109/CVPRW53098.2021.00380.

[43] Hu, L., Gao, L., Liu, Z., Feng, W.. Dynamic Spatial-Temporal Aggregation for Skeleton-Aware Sign Language Recognition. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Eval-*

ARTICLE IN PRESS

uation (LREC-COLING 2024), ELRA and ICCL, 5450–5460 (2024) Available at: https://aclanthology.org/2024.lrec-main.484/.

[44] Duy Khuat, B., Thai Phung, D., Thi Thu Pham, H., Ngoc Bui, A., Tung Ngo, S.. Vietnamese sign language detection using Mediapipe. In *Proceedings of the 2021 10th International Conference on Software and Computer Applications, Association for Computing Machinery*, 162–165 (2021) doi:10.1145/3457784.3457810.

[45] Nguyen, X., Nguyen, T., Tran, D., Bui, T., Nguyen, V.. An isolated Vietnamese Sign Language Recognition method using a fusion of Heatmap and Depth information based on Convolutional Neural Networks. In *2024 Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC), IEEE*, 1–6 (2024) doi:10.1109/APSIPAASC63619.2025.10848961.

[46] Tran, K. B., Nguyen, U. D., Huynh, Q. T.. Continuous Sign Language Recognition Using MediaPipe. In *2023 International Conference on Advanced Technologies for Communications (ATC)*, 493-498 (2023) doi:10.1109/ATC58710.2023.10318855.

[47] Barbhuiya, A. A., Karsh, R. K., Jain, R.. CNN based feature extraction and classification for sign language. *Multimedia Tools and Applications* ***80*** *(2)*, 3051–3069 (2021) doi:10.1007/s11042-020-09829-y.

[48] Lee, C., Ng, K. K., Chen, C., Lau, H., Chung, S., Tsoi, T.. American sign language recognition and training method with recurrent neural network. *Expert Systems with Applications* ***167***, 114403 (2021) doi:10.1016/j.eswa.2020.114403.

[49] Darmawan, I. D. M. B. A., Linawati, Sukadarmika, G., Wirastuti, N. M. A. E. D., Pulungan, R.. Indonesian sign language system (SIBI) dataset: Sentences enhanced by diverse facial expressions for total communication. *Data in Brief* ***60***, 111642 (2025) doi:10.1016/j.dib.2025.111642.

[50] Natarajan, B., Rajalakshmi, E., Elakkiya, R., Kotecha, K., Abraham, A., Gabralla, L. A., Subramaniyaswamy, V.. Development of an End-to-End Deep Learning Framework for Sign Language Recognition, Translation, and Video Generation. *IEEE Access* ***10***, 104358-104374 (2022) doi:10.1109/ACCESS.2022.3210543.

[51] Khan, A., Jin, S., Lee, G., Arzu, G. E., Minh Dang, L., Nguyen, T. N., Choi, W., Moon, H.. Deep Learning Approaches for Continuous Sign Language Recognition: A Comprehensive Review. *IEEE Access* ***13***, 55524-55544 (2025) doi:10.1109/ACCESS.2025.3554046.

Author Contributions

T.N.Q. conceptualized the study, developed the data set format, performed data collection, preprocessing, and experimental validation, and wrote the original manuscript. K.P.D. and V.T.D. performed data collection, annotation, and data curation. V.T.H. performed technical validation of the methodology. M.S., J.S., and S.B. performed formal analysis and the

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formal preparation/processing of the manuscript documentation. A.Ł., P.B., and R.M. provided methodology, project administration, supervision, and funding acquisition. All authors read and agreed to the final version of the manuscript.

Competing Interests

The authors declare no competing interests.

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Consent for Publication

Consent to publish the video files was obtained from all adults included in the study. For minors (below 18 years of age), consent to publish was also obtained from a parent or guardian. (Project Registration No. 2025-24/KQNC-CS).

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